

Onyx — PrinterKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **PrinterKit** (`SD:/lib/printerkit.so`, `user/Kits/printerkit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

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What it is

PrinterKit is printing: the Print dialog, a job and its pages — text, lines, shapes, pictures — handed to the print service, which makes of them what the printer takes, or a PDF file.

Include	<code>#include "printerkit/printerkit.h"</code>
Link	<code>lib/printerkit.imp.a</code>
Library	<code>SD:/lib/printerkit.so</code> — 33 entries in its table (<code>user/Kits/printerkit/printerkit.abi</code> , append-only)
Sources	<code>user/Kits/printerkit/</code>

Using it

A program prints in four steps: the Print dialog, a job, its pages, the end. Lengths are points (1/72 inch) from the page's top-left corner; colours are `0xRRGGBB`. The pages are recorded: the print service turns them into what the printer takes — or into a PDF file.

```
#include "printerkit/printerkit.h"
```

```
PrintSetup s;  
print_setup_default (&s);  
PrintDialogInfo di = { sizeof di, "My document", 1, 0, 0, 0, 0 };  
if (print_dialog (&s, &di)) // the printer, the paper, the  
pages  
{  
    PrintJob *j = print_begin (&s, "My document");  
    int font = print_font (j, "DejaVu Sans", 0);  
    if (print_page (j, 0, 0)) // 0, 0: the paper chosen  
    {  
        print_text (j, font, 12, 72, 84, "Hello, printer", 0x000000);  
        print_line (j, 72, 90, s.paper_w - 72, 90, 0.75f, 0x808080);  
    }  
}
```

```
    print_end (j); // to the print queue
}
```

Rectangles, paths (lines and curves, filled or stroked) and pictures are drawn the same way (`print_rect`, `print_path_*`, `print_image`).

Index

Everything the headers declare, in their order — the details are in each header's part below.

Name	What it does	Header
<code>PrintSetup</code>	What the user chose in the Print dialog.	<code>printerkit.h</code>
<code>PrintDialogInfo</code>	the page is fitted on the printer's paper	<code>printerkit.h</code>
<code>print_setup_default</code>	the default printer, its default paper, upright, one copy	<code>printerkit.h</code>
<code>print_dialog</code>	1: print (s filled), 0: cancelled	<code>printerkit.h</code>
<code>print_setup_paper</code>	another paper → 1, 0 unknown name	<code>printerkit.h</code>
<code>print_begin</code>	0: the spool cannot be written	<code>printerkit.h</code>
<code>print_page</code>	a new page (w, h 0: the setup's paper) → 1, or 0: outside	<code>printerkit.h</code>
<code>print_rect</code>	the pages to print -- skip its drawing (it would be ignored)	<code>printerkit.h</code>
<code>print_line</code>	the pages to print -- skip its drawing (it would be ignored)	<code>printerkit.h</code>
<code>print_frame</code>	a rectangle's outline	<code>printerkit.h</code>
<code>print_path_move</code>	a shape	<code>printerkit.h</code>
<code>print_path_line</code>	a shape	<code>printerkit.h</code>
<code>print_path_curve</code>	a shape	<code>printerkit.h</code>
<code>print_path_close</code>	a shape	<code>printerkit.h</code>
<code>print_path_fill</code>	a shape	<code>printerkit.h</code>
<code>print_path_stroke</code>	a shape	<code>printerkit.h</code>
<code>print_image</code>	an image	<code>printerkit.h</code>
<code>print_font</code>	text	<code>printerkit.h</code>
<code>print_text</code>	text	<code>printerkit.h</code>
<code>print_text_width</code>	text	<code>printerkit.h</code>
<code>print_font_metrics</code>	text	<code>printerkit.h</code>
<code>print_font_data</code>	for an app with its own	<code>printerkit.h</code>

Name	What it does	Header
	text layout	
<code>print_glyph</code>	for an app with its own text layout	<code>printerkit.h</code>
<code>print_end</code>	the job queued → its number (> 0), or < 0: it could not be	<code>printerkit.h</code>
<code>print_abort</code>	nothing printed	<code>printerkit.h</code>
<code>PrintPrinter</code>	(a type)	<code>printerkit.h</code>
<code>print_printers</code>	→ how many	<code>printerkit.h</code>
<code>print_printer_media</code>	a paper of a printer	<code>printerkit.h</code>
<code>print_printer_add</code>	1 / 0 (err: why)	<code>printerkit.h</code>
<code>print_printers_find</code>	the printers found on the network into out (max, 16 at most) → how many (0: none, no print service)	<code>printerkit.h</code>
<code>print_printer_remove</code>	taken off the list, by its name → 1 / 0 (not known; the PDF printer stays)	<code>printerkit.h</code>
<code>print_printer_default</code>	made the default printer, by its name → 1 / 0 (not known)	<code>printerkit.h</code>
<code>print_printer_status</code>	asked now: 1 ready (text: ink...), 0 (text: why)	<code>printerkit.h</code>
<code>print_jobs</code>	→ how many (the queue, then the last finished)	<code>printerkit.h</code>
<code>print_job_cancel</code>	a waiting or printing job stopped → 1 / 0 (no such job, no print service)	<code>printerkit.h</code>
<code>print_jobs_forget</code>	the finished jobs dropped from the list	<code>printerkit.h</code>

`printerkit/printerkit.h`

`printerkit/printerkit.h` -- printing from an app (the shared library `SD:/lib/printerkit.so`; docs/03 "Printing").

An app prints in four steps -- the Print dialog, a job, its pages, the end:

```
PrintSetup s; print_setup_default (&s);
PrintDialogInfo di = { sizeof di, "My document", pageCount, 0, 0, 0, 0 };
if (print_dialog (&s, &di)) // the user chose a printer,
a paper, the pages...
```

```

{
    PrintJob *j = print_begin (&s, "My document");
    int f = print_font (j, "DejaVu Sans", 0);
    for (int p = 0; p < pageCount; p++)
        if (print_page (j, 0, 0)) // 0, 0: the paper chosen; 0
returned: not in the range
    {
        print_text (j, f, 12, 72, 72 + 12, "Hello", 0x000000);
        print_line (j, 72, 90, s.paper_w - 72, 90, 0.75f, 0x808080);
    }
    print_end (j); // the job goes to printd's
queue
}

```

Lengths are points (1/72 inch), y goes down from the page's top-left corner, colours are 0xRRGGBB. The pages are recorded, not rendered: the print service (printd) makes of them what the printer takes -- a PDF file for the PDF printer, pages of pixels at its resolution for a network printer -- and tells the user when the job is done. The app does not wait for the printer.

A program links lib/printerkit.imp.a (with lib/uikit.imp.a and lib/fontkit.imp.a: the dialog is uikit's, the text FreeType's). The table is append-only (printerkit/printerkit.abi): these functions never change.

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```
typedef struct PrintJob PrintJob;
```

```

enum { PRINT_DRAFT = 3, PRINT_NORMAL = 4, PRINT_HIGH = 5 }; // quality
enum { PRINT_PORTRAIT = 0, PRINT_LANDSCAPE = 1 };
enum { PRINT_BOLD = 1, PRINT_ITALIC = 2 }; // a font's style
enum { PRINT_IMG_ALPHA = 1, // the pixels are 0xAARRGGBB (else 0x00RRGGBB:
opaque)
    PRINT_IMG_PHOTO = 2 }; // a photo: kept as a JPEG (smaller; not for text
or line art)

```

What the user chose in the Print dialog. Fields are only ever added at the end (size says how many).

```
typedef struct PrintSetup
```

```

{
    unsigned size; // sizeof (PrintSetup)
    char printer[64]; // the printer's name ("PDF": the PDF printer)
    char media[64]; // the paper, as IPP names it ("iso_a4_210x297mm")
    float paper_w, paper_h; // the page in points, the orientation applied
    float margin_l, margin_t, margin_r, margin_b; // what the printer cannot print
on, along each edge
    int orientation; // PRINT_PORTRAIT / PRINT_LANDSCAPE
    int copies;
    int mono; // 1: black and white
    int quality; // PRINT_DRAFT / NORMAL / HIGH
    int from, to; // the pages to print, from 1; 0, 0: all
    char output[256]; // the PDF printer: the file to write
}

```

```
int reserved[16];
} PrintSetup;
```

```
enum { PRINT_DLG_OWN_PAPER = 1 }; // the document has its own page size (a slide, a
PDF): no paper choice --
```

the page is fitted on the printer's paper

```
typedef struct PrintDialogInfo
{
    unsigned size; // sizeof (PrintDialogInfo)
    const char *title; // the document's name (the dialog's title, the PDF's
file name)
    int pages; // how many pages it has (0: not known)
    int current; // the page shown (from 1: offers "Current page"; 0:
no such choice)
    int flags; // PRINT_DLG_*
    float paper_w, paper_h; // the document's own paper to select first (0:
the printer's default)
} PrintDialogInfo;
```

the dialog

```
void print_setup_default (PrintSetup *s); // the default printer, its
default paper, upright, one copy
int print_dialog (PrintSetup *s, const PrintDialogInfo *info); // 1: print (s
filled), 0: cancelled
int print_setup_paper (PrintSetup *s, const char *media, int orientation); //
another paper -> 1, 0 unknown name
```

a job

```
PrintJob *print_begin (const PrintSetup *s, const char *title); // 0: the spool
cannot be written
int print_page (PrintJob *j, float w, float h); // a new page (w, h 0: the
setup's paper) -> 1, or 0: outside
```

the pages to print -- skip its drawing (it would be ignored)

```
void print_rect (PrintJob *j, float x, float y, float w, float h, unsigned rgb);
void print_line (PrintJob *j, float x0, float y0, float x1, float y1, float width,
unsigned rgb);
void print_frame (PrintJob *j, float x, float y, float w, float h, float width,
unsigned rgb); // a rectangle's outline
```

a shape: move to, lines and Bezier curves, closed or not; then filled, or stroked with a line

```
void print_path_move (PrintJob *j, float x, float y);
void print_path_line (PrintJob *j, float x, float y);
void print_path_curve (PrintJob *j, float x1, float y1, float x2, float y2, float
x3, float y3);
void print_path_close (PrintJob *j);
void print_path_fill (PrintJob *j, unsigned rgb);
void print_path_stroke (PrintJob *j, float width, unsigned rgb);
```

an image: pw x ph pixels drawn in (x, y, w, h)

```
void print_image (PrintJob *j, const unsigned *px, int pw, int ph, float x, float y,
float w, float h, int flags);
```

text: a font by its family ("DejaVu Sans", "Liberation Serif"...: SD:/res/fonts, SD:/fonts) and style → its number (-1: no such font nor a fallback); the text at its baseline → the pen's advance

```
int print_font (PrintJob *j, const char *family, int style);
float print_text (PrintJob *j, int font, float size, float x, float baseline, const
char *utf8, unsigned rgb);
float print_text_width (PrintJob *j, int font, float size, const char *utf8);
void print_font_metrics (PrintJob *j, int font, float size, float *ascent, float
*descent, float *line);
```

for an app with its own text layout: a TrueType file's bytes (kept by the app until print_end) → the font's number; one glyph of it at its pen position (style: PRINT_BOLD / ITALIC made from this face)

```
int print_font_data (PrintJob *j, const unsigned char *ttf, unsigned len);
void print_glyph (PrintJob *j, int font, float size, float x, float baseline,
unsigned glyph, unsigned unicode, unsigned rgb, int style);
int print_end (PrintJob *j); // the job queued -> its number (> 0), or < 0: it
could not be
void print_abort (PrintJob *j); // nothing printed
```

the printers and the queue (the Control Panel's applet; an app's own status)

```
typedef struct PrintPrinter
{
    char name[64], kind[8], uri[160], model[64]; // kind: "pdf", "ipp"
    int color, copies, dpi, is_default;
    unsigned quality; // bits 3 (draft), 4 (normal), 5 (high)
    char media_default[64];
} PrintPrinter;
enum { PRINT_JOB_QUEUED = 0, PRINT_JOB_PREPARING, PRINT_JOB_SENDING,
PRINT_JOB_PRINTING, PRINT_JOB_DONE, PRINT_JOB_FAILED, PRINT_JOB_CANCELED };
typedef struct PrintJobInfo { unsigned id; int state, pages, page; char printer[64],
title[80], message[96]; } PrintJobInfo;

int print_printers (PrintPrinter *out, int max); // -> how many
```

a paper of a printer: its IPP name, its name for a person → 1, 0: no i-th paper

```
int print_printer_media (const char *printer, int i, char *name, int ncap, char
*label, int lcap);
```

a network printer asked what it can do, and added (address: "192.168.0.14",
"ipp://host:631/ipp/print")

```
int print_printer_add (const char *name, const char *address, char *err, int cap);
// 1 / 0 (err: why)
```

the local network searched for printers (mDNS / DNS-SD, as AirPrint: a few seconds) → how many; usable 0: it answers but takes neither PWG Raster nor PDF

```
typedef struct PrintFound { char address[32], name[64], model[64]; int usable; }
PrintFound;
int print_printers_find (PrintFound *out, int max); // the printers found on the
network into out (max, 16 at most) -> how many (0: none, no print service)
int print_printer_remove (const char *name); // taken off the list, by its name ->
1 / 0 (not known; the PDF printer stays)
int print_printer_default (const char *name); // made the default printer, by its
name -> 1 / 0 (not known)
int print_printer_status (const char *name, char *text, int cap); // asked now: 1
ready (text: ink...), 0 (text: why)
int print_jobs (PrintJobInfo *out, int max); // -> how many (the queue,
then the last finished)
int print_job_cancel (unsigned id); // a waiting or printing job stopped -> 1 / 0
(no such job, no print service)
```

```
void print_jobs_forget (void);           // the finished jobs dropped from the
list

}
#endif
```